KITABU QUEST

P4

SCIENCE AND ELEMENTARY TECHNOLOGY

Chameleon

Cover scene

learners doing a simple science and technology investigation with a chameleon mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 4

RWA REB-CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST P4 Science and Elementary Technology

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P4 learners study Science and Elementary Technology one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Observe, ask questions, design safely, record evidence, and explain findings.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Objects production: Lesson Opener

Learning focus: Objects production

Country link: a safe Rwandan observation, model, data table, or investigation for Objects production.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- make useful and safe objects using local materials and take care of them properly
- identify common local materials used to make toys, utility, and learning objects
- explain the steps and techniques for making different objects using appropriate materials
- select the appropriate local materials based on the object to be made
- create toys, learning tools, and utility objects using correct techniques and materials
- show interest and curiosity in using local materials to create useful objects

Inquiry questions:
- How can objects production help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Objects production: Learn

Objects production is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Objects production
- Objects
- production
- concept
- observation
- evidence
- safety
- conclusion

Objects production: Worked Example

Investigation model for Objects production: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Objects production: Vocabulary And Study Skill

Use these words and study moves before you practise Objects production. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Objects production
- Objects
- production
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Objects production without a clear example, method, or evidence.

Objects production: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Objects production.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Objects production: Practice And Correction

Practice for Objects production:

Main outcome: make useful and safe objects using local materials and take care of them properly.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Objects production: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Objects production.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Objects production: Outcome Check

Outcome check for Objects production: Make useful and safe objects using local materials and take care of them properly.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Objects production: Outcome Check

Outcome check for Objects production: Identify common local materials used to make toys, utility, and learning objects.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

ICT basics: Lesson Opener

Learning focus: ICT basics

Country link: a safe Rwandan observation, model, data table, or investigation for ICT basics.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- use basic ICT terms correctly and manage files safely using Sugar Journal and a flash disk. Explain the five big ideas of AI and the characteristics of AI tools
- define ICT and its main components
- explain the roles of ICT in communication, education, business, and daily life
- explain common ICT terms and internet-related terminology
- use the Sugar Journal to manage activities
- organize and manage files in the Journal

Inquiry questions:
- How can ict basics help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

ICT basics: Learn

ICT basics is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- ICT basics
- basics
- concept
- observation
- evidence
- safety
- conclusion

ICT basics: Worked Example

Investigation model for ICT basics: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

ICT basics: Vocabulary And Study Skill

Use these words and study moves before you practise ICT basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- ICT basics
- basics
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about ICT basics without a clear example, method, or evidence.

ICT basics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for ICT basics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

ICT basics: Practice And Correction

Practice for ICT basics:

Main outcome: use basic ICT terms correctly and manage files safely using Sugar Journal and a flash disk. Explain the five big ideas of AI and the characteristics of AI tools.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

ICT basics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from ICT basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

ICT basics: Outcome Check

Outcome check for ICT basics: Use basic ICT terms correctly and manage files safely using Sugar Journal and a flash disk. Explain the five big ideas of AI and the characteristics of AI tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

ICT basics: Outcome Check

Outcome check for ICT basics: Define ICT and its main components.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphics and multimedia: Lesson Opener

Learning focus: Graphics and multimedia

Country link: a safe Rwandan observation, model, data table, or investigation for Graphics and multimedia.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- use basic graphic and multimedia tools to create simple drawings and open simple visual and audio content
- define Paint and explain its use
- describe steps to open, save, and close the Paint activity
- explain the use of common Paint tools and their properties
- open, save, and close a Paint activity on the computer
- control the movement of the touchpad (mouse) to interact with drawing tools

Inquiry questions:
- How can graphics and multimedia help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphics and multimedia: Learn

Graphics and multimedia is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Graphics and multimedia
- Graphics
- multimedia
- concept
- observation

Graphics and multimedia: Worked Example

Worked example for Graphics and multimedia: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Graphics and multimedia: Vocabulary And Study Skill

Use these words and study moves before you practise Graphics and multimedia. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Graphics and multimedia
- Graphics
- multimedia
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Graphics and multimedia without a clear example, method, or evidence.

Graphics and multimedia: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Graphics and multimedia.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Graphics and multimedia: Practice And Correction

Practice for Graphics and multimedia:

Main outcome: use basic graphic and multimedia tools to create simple drawings and open simple visual and audio content.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Graphics and multimedia: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Graphics and multimedia.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphics and multimedia: Outcome Check

Outcome check for Graphics and multimedia: Use basic graphic and multimedia tools to create simple drawings and open simple visual and audio content.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphics and multimedia: Outcome Check

Outcome check for Graphics and multimedia: Define Paint and explain its use.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Lesson Opener

Learning focus: Programming for children

Country link: a safe Rwandan observation, model, data table, or investigation for Programming for children.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- design and construct geometric shapes in Turtle art activity, create animations using Scratch programming and identify the main parts of a robot
- identify the main parts and tools in Turtle Art and Scratch windows
- explain how to open, save, and close projects in Turtle Art and Scratch
- describe how to draw shapes like squares, rectangles, and circles using Turtle Art
- open, save, and close activities in Turtle Art and Scratch
- navigate and identify the toolbars and components in both Turtle Art and Scratch

Inquiry questions:
- How can programming for children help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Learn

Programming for children is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Programming for children: Worked Example

Worked example for Programming for children: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Programming for children: Vocabulary And Study Skill

Use these words and study moves before you practise Programming for children. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Programming for children without a clear example, method, or evidence.

Programming for children: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Programming for children.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Programming for children: Practice And Correction

Practice for Programming for children:

Main outcome: design and construct geometric shapes in Turtle art activity, create animations using Scratch programming and identify the main parts of a robot.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Programming for children: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Programming for children.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Design and construct geometric shapes in Turtle art activity, create animations using Scratch programming and identify the main parts of a robot.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Identify the main parts and tools in Turtle Art and Scratch windows.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air, wind and sound: Lesson Opener

Learning focus: Air, wind and sound

Country link: a safe Rwandan observation, model, data table, or investigation for Air, wind and sound.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain the importance of air, wind, and sound, their uses and dangers, and learn how to protect yourself from loud noises and the environment
- state the composition of air
- explain the importance of air
- explain the uses and dangers of air components
- describe main properties of air

Inquiry questions:
- How can air, wind and sound help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air, wind and sound: Learn

Air, wind and sound is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Air, wind and sound
- wind
- sound
- concept
- observation
- evidence
- safety
- conclusion

Air, wind and sound: Worked Example

Investigation model for Air, wind and sound: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Air, wind and sound: Vocabulary And Study Skill

Use these words and study moves before you practise Air, wind and sound. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Air, wind and sound
- wind
- sound
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Air, wind and sound without a clear example, method, or evidence.

Air, wind and sound: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Air, wind and sound.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Air, wind and sound: Practice And Correction

Practice for Air, wind and sound:

Main outcome: explain the importance of air, wind, and sound, their uses and dangers, and learn how to protect yourself from loud noises and the environment.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Air, wind and sound: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Air, wind and sound.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air, wind and sound: Outcome Check

Outcome check for Air, wind and sound: Explain the importance of air, wind, and sound, their uses and dangers, and learn how to protect yourself from loud noises and the environment.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air, wind and sound: Outcome Check

Outcome check for Air, wind and sound: State the composition of air.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil: Lesson Opener

Learning focus: Soil

Country link: a safe Rwandan observation, model, data table, or investigation for Soil.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain what soil is, describe its uses and importance, and demonstrate care for protecting and using soil wisely
- identify types of soil
- describe the composition of soil
- explain how soil is used in daily life
- identify the characteristics of fertile soil

Inquiry questions:
- How can soil help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil: Learn

Soil explains soil as a living natural resource that supports plants, animals, and people.

Core idea:
- Soil is made from weathered rock, humus, air, water, and living organisms.
- Sandy, clay, and loam soils feel different, hold water differently, and suit different uses.
- Healthy soil supports plant roots, stores water, and provides nutrients.
- Soil can be damaged by erosion, burning, pollution, and careless farming.
- Learners can compare soil samples safely by colour, texture, water retention, and visible organic matter.

Key words:
- Soil
- concept
- observation
- evidence
- safety
- conclusion

Soil: Worked Example

Investigation model for Soil: Compare two soil samples safely.

1. Collect two small teacher-approved soil samples, such as garden soil and sandy soil.
2. Observe colour and texture without putting soil near the mouth or eyes.
3. Add a small amount of water and compare how each sample holds water.
4. Record results in a table: sample, colour, texture, water retention, visible organic matter.
5. Explain which sample may support plant growth better and why.

Conclusion: soil properties affect plant growth and land use.

Soil: Vocabulary And Study Skill

Use these words and study moves before you practise Soil. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Soil
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Soil without a clear example, method, or evidence.

Soil: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Soil.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Soil: Practice And Correction

Practice for Soil:

Main outcome: explain what soil is, describe its uses and importance, and demonstrate care for protecting and using soil wisely.

1. Compare two safe soil samples by colour, texture, and water retention.
2. Record observations in a table.
3. Explain which soil may support plant growth better and why.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Soil: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Soil.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil: Outcome Check

Outcome check for Soil: Explain what soil is, describe its uses and importance, and demonstrate care for protecting and using soil wisely.

1. Name two soil components.
2. Compare two soil properties.
3. Explain one way to protect soil from damage.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil: Outcome Check

Outcome check for Soil: Identify types of soil.

1. Name two soil components.
2. Compare two soil properties.
3. Explain one way to protect soil from damage.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Animals classification: Lesson Opener

Learning focus: Animals classification

Country link: a safe Rwandan observation, model, data table, or investigation for Animals classification.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- classify the animals according to how animals breathe, move, eat, and have babies and show care and curiosity about animals and their role in our world
- identify common animals based on their habitats and body coverings
- distinguish common animals based on their habitants and body coverings
- observe and describe the external features of different animals
- compare the outside features of animals to identify similarities and differences
- demonstrate care and responsibility in protecting animals and preserving wildlife

Inquiry questions:
- How can animals classification help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Animals classification: Learn

Animals classification is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Animals classification
- Animals
- classification
- concept
- observation
- evidence
- safety
- conclusion

Animals classification: Worked Example

Investigation model for Animals classification: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Animals classification: Vocabulary And Study Skill

Use these words and study moves before you practise Animals classification. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Animals classification
- Animals
- classification
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Animals classification without a clear example, method, or evidence.

Animals classification: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Animals classification.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Animals classification: Practice And Correction

Practice for Animals classification:

Main outcome: classify the animals according to how animals breathe, move, eat, and have babies and show care and curiosity about animals and their role in our world.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Animals classification: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Animals classification.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Animals classification: Outcome Check

Outcome check for Animals classification: Classify the animals according to how animals breathe, move, eat, and have babies and show care and curiosity about animals and their role in our world.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Animals classification: Outcome Check

Outcome check for Animals classification: Identify common animals based on their habitats and body coverings.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Rabbits and chickens management: Lesson Opener

Learning focus: Rabbits and chickens management

Country link: a safe Rwandan observation, model, data table, or investigation for Rabbits and chickens management.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe how to care for rabbits and chicken, choose healthy animals, build proper shelters, feed them correctly, protect them from diseases, show interest and care in raising them for home and community benefits
- describe the features of good rabbits and chicken to keep
- identify common diseases that affect rabbits and chicken
- identify the importance of rabbit farming
- choose good rabbits to rear
- design a small rabbit- keeping project

Inquiry questions:
- How can rabbits and chickens management help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Rabbits and chickens management: Learn

Rabbits and chickens management is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Rabbits and chickens management
- Rabbits
- chickens
- management
- concept
- observation
- evidence
- safety

Rabbits and chickens management: Worked Example

Investigation model for Rabbits and chickens management: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Rabbits and chickens management: Vocabulary And Study Skill

Use these words and study moves before you practise Rabbits and chickens management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Rabbits and chickens management
- Rabbits
- chickens
- management
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Rabbits and chickens management without a clear example, method, or evidence.

Rabbits and chickens management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Rabbits and chickens management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Rabbits and chickens management: Practice And Correction

Practice for Rabbits and chickens management:

Main outcome: describe how to care for rabbits and chicken, choose healthy animals, build proper shelters, feed them correctly, protect them from diseases, show interest and care in raising them for home and community benefits.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Rabbits and chickens management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Rabbits and chickens management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Rabbits and chickens management: Outcome Check

Outcome check for Rabbits and chickens management: Describe how to care for rabbits and chicken, choose healthy animals, build proper shelters, feed them correctly, protect them from diseases, show interest and care in raising them for home and community benefits.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Rabbits and chickens management: Outcome Check

Outcome check for Rabbits and chickens management: Describe the features of good rabbits and chicken to keep.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants germination: Lesson Opener

Learning focus: Plants germination

Country link: a safe Rwandan observation, model, data table, or investigation for Plants germination.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe how seeds grow into plants, identify plant parts and their roles, carry out simple planting activities, show care and curiosity about how plants grow and why they are important
- list the conditions of germination
- identify the main stages of germination
- identify the types of germination

Inquiry questions:
- How can plants germination help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants germination: Learn

Plants germination teaches useful computer work through exact steps, not guessing.

Core idea:
- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:
- Plants germination
- Plants
- germination
- concept
- observation
- evidence
- safety
- conclusion

Plants germination: Worked Example

Worked example for Plants germination: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Plants germination: Vocabulary And Study Skill

Use these words and study moves before you practise Plants germination. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Plants germination
- Plants
- germination
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Plants germination without a clear example, method, or evidence.

Plants germination: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Plants germination.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Plants germination: Practice And Correction

Practice for Plants germination:

Main outcome: describe how seeds grow into plants, identify plant parts and their roles, carry out simple planting activities, show care and curiosity about how plants grow and why they are important.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Plants germination: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Plants germination.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants germination: Outcome Check

Outcome check for Plants germination: Describe how seeds grow into plants, identify plant parts and their roles, carry out simple planting activities, show care and curiosity about how plants grow and why they are important.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants germination: Outcome Check

Outcome check for Plants germination: List the conditions of germination.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human sensory organs: Lesson Opener

Learning focus: Human sensory organs

Country link: a safe Rwandan observation, model, data table, or investigation for Human sensory organs.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- name the five sense organs, describe their functions, practice ways to care for them, and show concern for keeping their sense organs healthy and safe
- explain the function of ear
- describe the function of eyes
- state ways of keeping hygiene of sense organs
- practice keeping skin clean
- respond correctly when someone has a skin injury

Inquiry questions:
- How can human sensory organs help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human sensory organs: Learn

Human sensory organs is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Human sensory organs
- Human
- sensory
- organs
- concept
- observation
- evidence
- safety

Human sensory organs: Worked Example

Worked example for Human sensory organs: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Human sensory organs: Vocabulary And Study Skill

Use these words and study moves before you practise Human sensory organs. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Human sensory organs
- Human
- sensory
- organs
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human sensory organs without a clear example, method, or evidence.

Human sensory organs: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human sensory organs.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human sensory organs: Practice And Correction

Practice for Human sensory organs:

Main outcome: name the five sense organs, describe their functions, practice ways to care for them, and show concern for keeping their sense organs healthy and safe.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human sensory organs: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human sensory organs.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human sensory organs: Outcome Check

Outcome check for Human sensory organs: Name the five sense organs, describe their functions, practice ways to care for them, and show concern for keeping their sense organs healthy and safe.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human sensory organs: Outcome Check

Outcome check for Human sensory organs: Explain the function of ear.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human skeleton: Lesson Opener

Learning focus: Human skeleton

Country link: a safe Rwandan observation, model, data table, or investigation for Human skeleton.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- name the main parts of the skeleton, explain what it does, practice how to keep bones safe and healthy, and show care in preventing and responding to bone accidents
- identify the main parts of the skeleton
- identify the major bones of the legs and arms
- identify the main parts and big bones of the human skeleton
- draw and label the human skeleton
- give simple first aid when someone has a bone injury

Inquiry questions:
- How can human skeleton help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human skeleton: Learn

Human skeleton explains how muscles help the body move and stay active.

Core idea:
- Muscles pull on bones to create movement at joints.
- Muscles usually work in pairs: one contracts while the other relaxes.
- Exercise, rest, clean food, and safe posture help muscles remain healthy.
- Sudden heavy work, poor warm-up, and unsafe movement can injure muscles.
- A labelled body diagram helps learners connect muscle action with daily movement.

Key words:
- Human skeleton
- Human
- skeleton
- concept
- observation
- evidence
- safety
- conclusion

Human skeleton: Worked Example

Model for Human skeleton: Explain how the arm bends.

1. Hold one arm out and bend it slowly.
2. Feel the upper-arm muscle shorten as the elbow bends.
3. Straighten the arm and notice the movement changes.
4. Draw a simple arm diagram with bone, joint, and muscle labels.
5. Add one safety note about warm-up or avoiding sudden heavy lifting.

Conclusion: muscles pull on bones at joints to make movement.

Human skeleton: Vocabulary And Study Skill

Use these words and study moves before you practise Human skeleton. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Human skeleton
- Human
- skeleton
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human skeleton without a clear example, method, or evidence.

Human skeleton: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human skeleton.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human skeleton: Practice And Correction

Practice for Human skeleton:

Main outcome: name the main parts of the skeleton, explain what it does, practice how to keep bones safe and healthy, and show care in preventing and responding to bone accidents.

1. Draw and label a simple joint and muscle action.
2. Explain how a muscle helps movement by pulling.
3. List two habits that protect muscles during work or exercise.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human skeleton: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human skeleton.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human skeleton: Outcome Check

Outcome check for Human skeleton: Name the main parts of the skeleton, explain what it does, practice how to keep bones safe and healthy, and show care in preventing and responding to bone accidents.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human skeleton: Outcome Check

Outcome check for Human skeleton: Identify the main parts of the skeleton.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Muscles: Lesson Opener

Learning focus: Muscles

Country link: a safe Rwandan observation, model, data table, or investigation for Muscles.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- name the main types of muscles, explain what muscles do, practice safe exercises to keep muscles strong and show care in protecting their muscles from injuries
- investigate and explain muscles using evidence
- list the main voluntary and involuntary muscles in the body
- explain the function of voluntary muscles
- use simple ways to keep muscles strong and healthy
- give basic first aid when someone has a muscle injury

Inquiry questions:
- How can muscles help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Muscles: Learn

Muscles explains how muscles help the body move and stay active.

Core idea:
- Muscles pull on bones to create movement at joints.
- Muscles usually work in pairs: one contracts while the other relaxes.
- Exercise, rest, clean food, and safe posture help muscles remain healthy.
- Sudden heavy work, poor warm-up, and unsafe movement can injure muscles.
- A labelled body diagram helps learners connect muscle action with daily movement.

Key words:
- Muscles
- concept
- observation
- evidence
- safety
- conclusion

Muscles: Worked Example

Model for Muscles: Explain how the arm bends.

1. Hold one arm out and bend it slowly.
2. Feel the upper-arm muscle shorten as the elbow bends.
3. Straighten the arm and notice the movement changes.
4. Draw a simple arm diagram with bone, joint, and muscle labels.
5. Add one safety note about warm-up or avoiding sudden heavy lifting.

Conclusion: muscles pull on bones at joints to make movement.

Muscles: Vocabulary And Study Skill

Use these words and study moves before you practise Muscles. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Muscles
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Muscles without a clear example, method, or evidence.

Muscles: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Muscles.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Muscles: Practice And Correction

Practice for Muscles:

Main outcome: name the main types of muscles, explain what muscles do, practice safe exercises to keep muscles strong and show care in protecting their muscles from injuries.

1. Draw and label a simple joint and muscle action.
2. Explain how a muscle helps movement by pulling.
3. List two habits that protect muscles during work or exercise.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Muscles: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Muscles.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Muscles: Outcome Check

Outcome check for Muscles: Name the main types of muscles, explain what muscles do, practice safe exercises to keep muscles strong and show care in protecting their muscles from injuries.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Muscles: Outcome Check

Outcome check for Muscles: Investigate and explain muscles using evidence.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Objects production: Review Clinic 1

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Objects production that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

ICT basics: Review Clinic 2

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from ICT basics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Graphics and multimedia: Review Clinic 3

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Graphics and multimedia that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Programming for children: Review Clinic 4

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Programming for children that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Science and Elementary Technology.

- Objects production: Explain this term using an example from the book, your school, or your community.
- ICT basics: Explain this term using an example from the book, your school, or your community.
- Graphics and multimedia: Explain this term using an example from the book, your school, or your community.
- Programming for children: Explain this term using an example from the book, your school, or your community.
- Air, wind and sound: Explain this term using an example from the book, your school, or your community.
- Soil: Explain this term using an example from the book, your school, or your community.
- Animals classification: Explain this term using an example from the book, your school, or your community.
- Rabbits and chickens management: Explain this term using an example from the book, your school, or your community.
- Plants germination: Explain this term using an example from the book, your school, or your community.
- Human sensory organs: Explain this term using an example from the book, your school, or your community.
- Human skeleton: Explain this term using an example from the book, your school, or your community.
- Muscles: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Science and Elementary Technology answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Science and Elementary Technology that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.